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Taxation and educational development: Evidence from British India

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ABSTRACT

This paper measures the effects of public expenditures on literacy in early 20th century British India. Using a new dataset and an instrumental variables strategy, I find that public investments in primary education had positive and statistically significant effects on literacy. A 10 percent increase in 1911 per-capita spending or 44 additional primary schools would have translated into a 2.6 percentage point increase in 1921 literacy in the population aged 15–20. The findings, however, differ by gender: the IV estimates on spending are statistically significant only for male literacy. India's historical experience thus suggests that building more schools would not have solved the problem of female illiteracy that continues to persist even today.

1. Introduction

Do public educational investments affect outcomes such as test scores, enrollment, literacy and wages?[1] Several researchers have studied this question in the context of US school finance reforms (Hoxby, 2001), government mid-day meal programs in India (Misra and Behera, 2003) and school construction projects in Indonesia (Duflo, 2001). The estimates in the literature, however, mirror the wide geographic coverage of the papers ranging from negative to large positive effects of spending.[2] A key reason for the range of estimates across studies is differences in the treatment of demand side factors.

To measure the causal effect of public investments, we need to understand how expenditures interact with other factors that influence schooling. A child (or adult) invests in education if the individual benefits (for example, higher wages) exceed the costs (for example, school fees). Family background such as parental income and education often influence the decision by altering benefits and costs. Given the social returns to education, states can also play an important role by providing education when the private market under-supplies it either because of low private returns or because of inaccurate information on returns. Public schools can have a large impact in places where families are credit constrained or where intergenerational commitment problems between children and parents lead to sub-optimal choices. Disentangling public expenditures from endogenous variables, however, is the key empirical challenge since higher spending may be a partial response to private demand.

In this paper, I estimate the effects of colonial public investments on literacy in early 20th century British India.[3] This is an important setting for three reasons. First, although India was the largest colony of the British Empire, our knowledge of colonial education is modest. We know public spending was low relative to world standards and other countries that faced similar

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[1] Hanushek (1986) and Glewwe (2002) review the US and developing country literature, respectively.

[2] Developing country studies suggest that estimates range from negative to large positive effects of the order of 2 standard deviation increases in test scores (Glewwe, 2002). The latter are from the introduction of blackboards and school roof repairs in Ghana. The impact of teacher quality in India has been positive but small (Kingdon, 1996).

[3] British India refers to approximately two-thirds of the Indian sub-continent that was under direct colonial control. The remaining one-third of the territories (Princely States) was under the rule of native kings who deferred to the British with regard to defense and foreign policy, but managed their own local affairs. See Fig. 1 – Map of India.

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returns (Davis and Huttenback, 1986), but we do not know whether low spending was a response to a perceived inefficacy of expenditures, budget constraints or conflicting priorities.[4] Given the historical roots of India's current service and education based economic growth (Broadberry and Gupta, 2010), understanding the response of literacy to public spending has important implications for India's growth path in the colonial and post-independence period.

Second, public expenditures vary dramatically within India, making it possible to identify the effects of spending. Because of historical circumstances, provinces such as Bombay had high public investments per-capita whereas other provinces such as Bengal, Bihar and Orissa received relatively low public revenues and consequently spent less on education. Within provinces land tax revenues contributed to differences in public spending, especially on primary education, because additional surcharges on land revenues basically funded district boards responsible for rural primary schooling. Although board expenditures do not capture all public investments, they represent most of the spending on public primary education and are commonly reported in the historical sources. Using board expenditures, this paper is the first to rigorously study the historical relationship between spending and literacy in India.

Third, differences in literacy across modern Indian states are remarkably similar to those in the historical period. For example, in 1991 the western states of Gujarat and Maharashtra had over 60 percent literacy as compared to 38 percent in the eastern state of Bihar. Although literacy was lower in the early 1900s, the regional differences are comparable: literacy in Bombay Presidency, which roughly represents contemporary Gujarat and Maharashtra, was almost twice as high as in Bihar and Orissa (11 percent versus 5 percent in 1931). How do we account for the initial variation in outcomes? Did public investments contribute to the differentials? Only by answering these questions can we begin to understand the persistence of the regional patterns into the 1990s.[5]

For the empirical exercise, I constructed a new dataset linking district-level information on public expenditures from the Indian district gazetteers to estimates of literacy reported in the censuses of 1911 and 1921. Controlling for province fixed effects and socio-economic factors, the OLS estimates find a positive and statistically significant effect of district board expenditures on both male and female literacy. Since the OLS estimates may be endogenous, I also use land revenues as an instrument for spending. According to historical accounts, subjective forces unrelated to education created quasi-random variation in revenue assessments conditional on observable differences in geography and development across districts. Moreover, land revenues are statistically unrelated to the first available measures of agricultural productivity (post-independence), corroborating the historical accounts and providing some reassurance about the validity of the approach.

In the IV specifications, public expenditures lead to higher literacy but the magnitude of the coefficients is smaller: a 10 percent increase in 1911 per-capita spending, or 44 additional primary schools, translates into a 2.6 percentage point increase in 1921 literacy in the population aged 15–20 as compared to a 3.9 percentage point increase under OLS. The OLS and IV estimates also differ by gender. Although the OLS estimates find significant effects for males and females, the IV estimates are statistically significant only for male literacy, indicating that colonial investments perhaps did not lead to better outcomes for women.

To test the validity of the instrument, I perform several robustness checks. First, as a control outcome I use 1921 cohorts that were unexposed to 1911 public spending either because they were too old or too young to be in school. Second, I use English literacy as another control outcome because the boards allocated money to vernacular primary schools only and so in principle they should not positively affect English literacy. The IV estimates are statistically uncorrelated both with the younger and older cohort-specific literacy, and with English literacy. Third, I add controls for different land tenure systems and public expenditures on non-educational services that could perhaps be related to land revenues and any omitted variables correlated with literacy. Fourth, I explore the effects of changes in spending between 1901 and 1911 on the production of literates, namely the change in literacy between 1921 and 1911. Akin to a quasi-first difference estimation, this effectively controls for district fixed effects and also finds positive effects of public spending on male literacy.

More public money would have increased male literacy in British India, but money alone would not have solved the problem of female illiteracy that continues to persist today. India's historical experience thus suggests that alleviating contemporary female illiteracy requires a more nuanced policy response cognizant of the cultural and social barriers to female education. The remainder of the paper is organized as follows: the next section provides a brief overview of educational expenditures in the colonial period; Section 3 lays out the empirical strategy; Section 4 describes the district-level data; Section 5 discusses the results and Section 6 concludes.

2. District board expenditures and land revenues in India

This section discusses colonial investments in primary education and argues that taxes on land (or land revenues) provide a plausibly exogenous source of variation to identify the effects of public expenditures on outcomes.

During the colonial period, the East India Company and the British Crown introduced a new state system of education in the 19th century that largely replaced the former indigenous system.[6] From 1858 to 1919, the Crown via the Government of

[4] Whitehead (2005) reviews the limited quantitative literature on colonial Indian education.

[5] Development economists have extensively discussed the contemporary regional variation in outcomes within India. See Roy and Datta (1993), Drèze and Sen (1998), Probe Team (1999), Drèze and Kingdon (2001), Pratham (2006), Kingdon (2007), and Pal and Ghosh (2007).

[6] Under the former indigenous system there were elite religious schools serving a small number of students interested in an academic career and local elementary schools to meet the more mundane needs of the village community. See Nurullah and Naik (1951) and Basu (1982) for details.

India directly controlled education policy although the public financing and management of schools was decentralized to provincial governments beginning in the 1870s. Primary education and other local services such as infrastructure were further decentralized to rural district and urban municipal boards in the 1880s.[7] Many districts fail to report spending by provincial governments and urban boards in the historical sources and consequently, I use district board spending as the measure of public investments.

Although the boards capture the majority of public spending on rural primary education, they may also introduce certain biases in the estimation. The top panel of Table 1 reports district board expenditures as a percentage of total educational spending, primary school spending and public primary school spending in 1912–1913 for British India and for the larger provinces. On average, district councils accounted for 15 percent of total spending, 49 percent of primary education spending and 65 percent of public spending on primary education.[8] Since board spending underestimates total public spending, using these figures will generate a downward bias on the estimates because they measure public spending with noise.

Private spending could also interact with district board spending to either underestimate or overestimate the effects of public spending. If total spending is more evenly distributed than board spending, using districts boards will lead to a downward bias on the effects of public spending. However, if total spending is less evenly distributed, then there will be an upward bias on the effects of public spending. In Table 1, there is a small positive correlation across provinces between board spending and the proportion of total spending on primary education, suggesting that total spending was perhaps more evenly distributed than board spending. Using board spending should thus provide a lower bound on the effects of public spending.

Average spending by district boards was £3.5 per 1000 people, representing less than 0.5 percent of per-capita GDP in 1912–1913.[9] Public spending did increase over time, but always remained under 1 percent of per-capita GDP until Indian independence.[10] Besides the low level, Table 1 also highlights the significant heterogeneity within India, with Bombay outspending Bihar and Orissa by a factor of 10.[11] Spending also varied across districts within the same province: for example, spending per-capita ranged from Rs. 0.015 to 0.049 in Bihar and Orissa.

The bottom panel of Table 1 documents the relationship between spending and revenues. In the 19th century, additional surcharges on land revenues known as cesses represented 60 percent of district board revenues. Provincial governments, however, began making larger grants to boards in the early 1900s and by 1929–1930 grants contributed 43 percent to income as compared to 36 percent from cesses.[12] Cesses and grants jointly accounted for 75 percent of board revenues, while the rest came from tolls, school fees and other miscellaneous sources.

Grants are included under ‘other sources’ in the bottom panel of Table 1 and land revenues were primarily responsible for the observed differences in grants across provinces. In Bengal and Bihar, land revenue was fixed in cash in perpetuity because of the Permanent Settlement of 1793.[13] Consequently, these provinces were unable to recoup higher revenues as agricultural productivity and prices increased over the next century and they lost more public money when land revenues were decentralized to provinces in the late 19th century. Because of their low revenues, they transferred smaller grants to their districts.[14] Land revenues were revised every 30 years in Temporary Settlement areas such as Bombay and Madras, and accounted for changes in agricultural conditions. They had larger public funds and thus, also had more money to spend on grants.[15]

Provinces allocated grants to districts on a per-capita basis and sometimes favored either poorer (‘needy’) districts or those with larger minority populations.[16] Although land revenues did not influence grants within provinces, they contributed to differences in cesses both across and within provinces because cesses were additional taxes on land revenues. Cesses were usually levied at the rate of 6.25 percent on land revenues and the rate was uniform for districts within a province.[17]

[7] Rural district boards were set up by Lord Ripon in 1882 and were primarily responsible for the management of primary education, local infrastructure and medical services in rural areas. See Chand (1947) and Tinker (1968) for details.

[8] The estimate of district board spending as a proportion of total public spending on primary education is biased downwards because of differences in reporting years between the historical sources. Public primary education expenditures are reported for 1916-17 (Progress of Education in India, 1923), while the data on district board spending is from 1912–1913 (Statistical Abstracts of India). The Statistical Abstracts also severely undercount the extent of district board spending in the southern province of Madras since they appear to have excluded spending by sub-district boards (taluk boards) that were in fact a part of district boards. For the empirical analysis, I use district-level data obtained from the Indian District Gazetteers that report total spending by district councils including the lower taluk boards.

[9] This holds for the range of GDP estimates reported in Kumar (1982).

[10] In fact, public investments in human capital in British India were among the lowest in the world (Davis and Huttenback, 1986). Comparable areas such as the Indian Princely States spent twice as much on education and other foreign underdeveloped countries spent five times as much.

[11] The differential patterns in spending are not peculiar to 1912–1913 and hold for most of the colonial period. See Kumar (1982) and Statistical Abstracts of India.

[12] Calculations based on Chand (1947). The grants were targeted at rural primary education in particular.

[13] The Permanent Settlement of 1793 was a contract between the East India Company and the landlords of Bengal and Bihar. See Kumar (1982) for details.

[14] Permanent Settlement districts were also correlated with the *zamindari* system of land tenure whereby landlords were made responsible for revenue payments; in Temporary Settlement districts individual cultivators or village bodies were responsible for these payments. See Baden-Powell (1907, 1972) and Banerjee and Iyer (2005) for details on different land tenure arrangements.

[15] Land revenues in Bombay and Madras were assessed at higher rates than the other Temporary Settlement provinces and hence they had some of the highest spending during the colonial period. See Chand (1930, 1931), Misra (1960) and Kumar (1982) for details of the colonial fiscal system.

[16] See Quinquennial Review of Education (1902–1907). Chaudhary (2009) finds a positive correlation between the number of primary schools directly managed by local boards and the population share of minorities such as lower castes and tribes over the 1901–1911 decade. According to Chand (1947) some of the earlier grants were distributed in accordance with the amount of money a district raised (i.e. cess), but over time the rhetoric favored giving more money to needy districts. Some provinces did not have an official policy on how grants were distributed to districts, but the empirical analysis in Chaudhary (2009) supports the idea that minority districts did receive more money on average to set up schools.

[17] For example, if a district owed Rs. 30,000 in land revenues, the cess amount would equal Rs. 1875 ($0.0625 * 30{,}000$).

Table 1
District board educational expenditures and income.

	DB educational expenditures in 1912–1913			
	Per 1000 of the population (£)	As % of total educ. exp.	As % of total primary educ. exp	As % of total public primary educ. exp.
British India	3.59	15	49	65
Assam	5.60	30	84	87
Bengal	2.43	8	40	80
Bihar and Orissa	1.65	13	34	61
Bombay	10.27	20	42	59
Central provinces and Berar.	4.58	25	61	66
Madras	3.03	12	34	38
Punjab	5.12	18	85	93
United provinces	3.55	21	93	100
	DB revenues in 1912–1913 per 1000 of the population(£)			
	Cesses (additional taxes on land revenues)	Other sources (for e.g. provincial grants)	Total income	% of Cesses and other sources to total income
British India	6.53	5.96	16.65	75
Assam	7.40	7.02	23.11	62
Bengal	4.28	4.35	10.51	82
Bihar and Orissa	4.56	3.08	9.66	79
Bombay	10.84	10.23	25.78	82
Central provinces and Berar	4.99	9.34	16.07	89
Madras	10.80	10.13	31.85	66
Punjab	10.05	6.75	22.92	73
United provinces	5.20	4.26	11.53	82

Source: Statistical abstracts of india and progress of education in India, Eighth Quinquennial Review (1917–1922). The statistical abstracts separate educational expenditures into direct expenditures on different levels of education and indirect expenditures on buildings, etc. I constructed total primary education expenditure as the sum of direct expenditures on primary schools and a proportion of indirect expenditures on buildings with the proportion equal to the direct primary school expenditures relative to total direct expenditures on education. Information on public primary education expenditures is from 1916 to 1917 and was obtained from the Quinquennial Review, Volume II, Supplemental Tables, pp. 125. In the bottom panel, provincial contributions or grants to district boards are included under "other sources" along with school fees and miscellaneous income sources. The omitted category of income is tolls.

Since land revenues had been fixed in Permanent Settlement districts in 1793, the cess assessment was based on the annual value defined as the "rent paid by the tenant to the landlord" (Chand, 1947, pp. 118). Surveys were conducted to measure annual values, but were often outdated and inaccurate. In practice the cesses in Permanent Settlement districts appear to be based on the notoriously inaccurate land revenue assessments of 1793 because of the high correlation between 1911 per-capita cesses and land revenues (0.9).[18] Since cesses were based on idiosyncratic land assessments conducted in 1793, they should be unrelated to factors that may influence the demand for education almost 150 years later.

In Temporary Settlement districts, the cesses were assessed on land revenues calculated as a proportion (roughly 50 percent) of monetized net agricultural produce in Madras, a function of general conditions adjusted for different soil types in Bombay and roughly 50 percent of rental assets in the United Provinces and Punjab.[19] Revenue rates were high in the early to mid-19th century but declined over the late 19th and 20th century and represented less than 5 percent of net agricultural produce by the 1930s (Roy, 2000 ; Kumar, 1982).[20] The assessments were revised every 30 years and so were relatively insensitive to changes in agriculture, development and population. Detailed cadastral surveys were conducted at each reassessment, but subjective forces influenced the calculation with the system "lacking in precision and objectivity."[21] Soil, rainfall and a general level of development were some objective measures that entered into the calculations.

Although Indian economic historians have frequently linked revenue assessments to Indian agricultural distress, they have devoted less attention to the general inaccuracy of revenues – a topic frequently discussed by contemporaries and

[18] See Chand (1947) for details on cesses in Bengal. In particular, land revenue assessments in Permanent Settlement Bengal were not based on "any area survey, any consideration, that is, of the number, various fertility, or productive power, of the acres held in each case, or of the influence of proximity to markets and facility of communication, on the value of produce" (Baden-Powell, 1972, vol.1, pp. 287). They were largely a function of historical revenues and negotiations between Indian landlords and the East India Company. See Baden-Powell (1972) and Kumar (1982) for details.

[19] The cess was assessed on double the land revenue in the United Provinces and Punjab. See Baden-Powell (1972) and Chand (1931) for details on the assessment of land revenues in each province. Kumar (1982) highlights the random heterogeneity of the initial 19th century revenue assessment in Madras.

[20] Tax rates were generally the same within provinces and so differences in land assessments were the main reason for differences in land revenues across districts in the same province.

[21] Chand (1947), pp. 119. The surveys involved demarcation of boundaries, a record of land holdings and rights, and an assessment of the soil conditions. See Baden-Powell (1907, 1972) and Chand (1931, 1947).

subsequent researchers.[22] Settlement officers set the revenue demand at each reassessment and the ''idiosyncrasies of the Settlement Officers'' led to substantial quasi-random variation in land revenues (Chand, 1931, 1947). For example, ''according to Mr. Mackee, the Settlement Commissioner and Director of Land Records, the pitch of assessment varies in the most unreasonable fashion even from group to group in the same taluka and from district to district, so that the assessment may be half the rent in one place and only one-fifth in another'' (Chand, 1931, pp. 61). Baden-Powell's detailed description of the land settlement process also acknowledges the role of ''intuitive calculation'' and states ''with all these different methods, it is apt to be supported that, after all, Settlement is very much a matter of individual taste and opinion'' (Baden-Powell, 1972, vol. I, pp. 338) while defending the assessment of land revenues.

Chand (1931) highlights problems of assessments based on general considerations as in Bombay, but even in United Provinces where rental values formed the basis of assessment, the computation was often random because rents were not readily observed and ''there must necessarily be a point where estimation – guesswork if the term is preferred – comes in'' (Baden-Powell, 1972, vol. I, pp. 339).[23] Even when rents were directly observed, they only captured the value at the time of the settlement, with subsequent rents becoming ''largely matters of agreement between landlord and tenants'' (Stokes in Kumar, 1982).

Some objective factors obviously influenced the assessment. Soil, rainfall and proximity to railways are common variables that appear in historical discussions.[24] Anecdotal evidence in Baden-Powell (1972) and Chand (1931), however, suggests that schools and education never appear in the discussions. Land revenues may thus offer an instrument for board expenditures because they are correlated with revenues (i.e. cesses) and may be uncorrelated with unobservable returns to literacy controlling for differences in geography and socio-economic factors. Section 3 provides further evidence on the potential randomness of land revenues by exploring the relationship between revenues and agricultural productivity.

Colonial policy thus created differences in public investments in primary education within British India. The land settlements (Temporary versus Permanent) led to extreme disparity in inter-provincial patterns of public spending (via grants), whereas the idiosyncratic assessment of land revenues created quasi-random variation in public spending (via cesses) across districts in the same province. The empirical analysis exploits both the cross-sectional and temporal differences in spending to identify the effects on literacy.

3. Empirical framework

Although public expenditures accounted for 50 percent of total spending on primary education in British India, no study has evaluated whether higher spending in the colonial period led to better outcomes. In fact, British administrators took a dim view of the efficacy of public investments and official rhetoric often emphasized demand-side issues as the primary driver of educational outcomes (Nurullah and Naik, 1951).[25] Their position may partially reflect an official strategy to absolve the Imperial Government from any blame for the low level of investment. At the other extreme Congress leaders and Indian nationalists bemoaned the low level of public spending and strongly advocated higher expenditures as the key to better outcomes.[26]

Disagreement on the effects of educational expenditures in colonial India mirrors the contemporary disagreement on the causal effect of expenditures both in the US literature and in studies focusing on developing countries. Identifying the causal effects is difficult because of confounding factors such as income and development that are correlated with both investments and outcomes. According to economic theory, an individual uses costs and benefits as the basis of the decision to invest in literacy. Both individual characteristics such as ability, family background, parental education, social and religious affiliation, and community characteristics such as economic conditions and public educational investments influence the decision-making process. District level outcomes are an aggregate measure of the underlying individual decisions assuming every individual who invests in education becomes literate. In the absence of historical information on test scores or enrollment, I focus on literacy as the outcome variable because it is a reliable measure of the efficacy of public expenditures and because historical data are available on total and cohort-specific literacy.

As discussed earlier, colonial board expenditures were primarily determined by cesses and grants although public school fees also contributed a small amount to revenues. A significant portion of inter-provincial differences in expenditures was due to historical circumstances, i.e. the type of land settlement, but the analysis does not exploit difference across provinces because Indian provinces differ along many dimensions above and beyond the type of land settlement. The econometric analysis therefore uses variation across districts in the same province by including province dummies. A traditional OLS estimation nonetheless is subject to endogeneity problems.

[22] The discussion in Chand (1931) suggests that the marked diversity and inaccuracy of the system was realized only after official evidence was collected by the Taxation Enquiry Committee of 1924–1925, which highlighted that practices on the ground were very subjective relative to theory. Chand (1931) reviews the revenue assessment basis in each province and emphasizes the general inaccuracy and randomness of the system. Baden-Powell (1972) reviews the revenue assessment process in each province and, although he defends the system, his discussion also suggests that Settlement officers along with objective factors such as soil, etc. determined the final assessment.

[23] Chand (1931) also notes that ''much depends upon the personal judgment of the officers in charge of the work of assessment or reassessment'' (pp. 60).

[24] See Baden-Powell (1972) for details. He highlights that soil classification was a very important criterion for assessment in all provinces other than United Provinces and Punjab where rents formed the basis of assessment.

[25] See Progress of Education in India, Quinquennial Reviews, Government of India (1886–1937) that highlight official views on educational expenditures with statements such as – there is little to be said for opening schools to which parents will not send their children|| (Eighth Quinquennial Review, pp. 123) without compulsory education or greater demand.

[26] See Quinquennial Reviews (1886–1937) that reference Gokhale and other Indian nationalists.

First, if literacy affects public spending, then we cannot infer whether spending leads to literacy or literacy leads to spending, a problem of reverse causality. We can, however, address the issue by using lagged values of expenditures. If we assume that a child generally attends primary school between the ages of 5 and 10 where he acquires basic literacy skills, then a primary school pupil in 1911 would on average fall in the cohort aged 15–20 in 1921. Consequently, we can examine the effect of 1911 expenditures on 1921 literacy in the population aged 15–20.

Second, we have to control for variables that affect both public spending and literacy. For example, districts with higher population density (i.e. more urbanized) may have a higher private demand for education and more public resources to supply schools. Omitting population density would thus generate an upward bias on the OLS coefficient. To address the problem of omitted variables, I include many variables to capture differences across districts in the costs and benefits of literacy. Since parental education is a critical input into an individual's schooling decision, I control for the share of the population supported by professionals (doctors, lawyers and so forth), the best available measure reported in the historical sources. But professionals are perhaps a lower bound on the family background effect because it limits the educated parental population to professionals. I also introduce variables to capture the share of the population supported by commerce and industry because economic structure can also influence schooling. The opportunity cost of the time of a rural child who worked in the field with his agricultural parents was perhaps higher than that of a child supported by commerce or by industry.[27]

Because of the high degree of social heterogeneity in India and the potential variation in demand for schooling across different groups, I control for the population share of Brahmans (the traditional educated caste of Hindus), Muslims, Christians, Buddhists, and minorities such as lower castes and tribes. I also include a measure of caste and religious fragmentation, which has a strong negative effect on the supply of private primary schools in the colonial period (Chaudhary, 2009) and could potentially affect the demand for education. In addition, I control for population density, access to railways and income to capture differences in economic conditions.

In spite of the controls and province fixed effects, the OLS estimates may still pick up some measure of unobservable heterogeneity across districts. Land revenues, however, offer an instrumental variable because they were strongly correlated with board expenditures and were unlikely to be systematically related to the error term. Anecdotal evidence indicates that revenue calculations were plagued with idiosyncrasies creating quasi-random variation in both land revenues and cesses. Land revenues determined cesses and so we expect them to strongly influence board spending.

Table 2 examines the determinants of 1911 per-capita board expenditures on education, and land revenues is a key explanatory variable. Although the coefficient is higher in specifications without province fixed effects, almost two-thirds of the land revenue effect is driven by differences within provinces. A 10 percent increase in per-capita revenues leads to a 3 percent increase in average educational expenditures (specification 7) with an F-statistic of 12 on land revenues. The dummy for Temporary Settlement districts is significant in the across province comparisons (specifications 3 and 4), but neither this variable nor the type of land tenure system (proportion of non-landlord tenures) from Banerjee and Iyer (2005) has any statistically significant effect on expenditures controlling for province fixed effects (specification 5). This is because land settlements and land tenure systems did not vary within provinces.

Land revenues are thus strongly correlated with spending, but a valid instrument also has to be unrelated to unobservable returns to literacy. In principle, the revenues were supposed to capture a percentage of the rental assets or agricultural surplus of a district. Soil and rainfall were two objective factors that influenced the assessment computation. For example, districts with alluvial soil enjoyed a larger agricultural surplus and probably paid higher land taxes. Nonetheless, discussions on assessment procedures (Baden-Powell, 1972; Chand, 1931) indicate that revenues varied idiosyncratically even after accounting for differences in geography and development.

Unfortunately, there are no reliable district measures of actual agricultural productivity in the colonial period to assess the randomness of land revenues. The India Agriculture and Climate Data Set (IACDS) reports the first available measures for the years 1957–1987. Using these data, Table 3 explores the correlation between productivity in the post-independence period and colonial land revenues, controlling for differences in geography, rainfall and socio-economic factors (i.e., the same set of controls as in Table 2).[28] Columns one to four focus on agricultural productivity of the major crops, namely bajra, jowar, maize, rice and wheat, while columns five to eight focus on all crops reported in the IACDS.[29]

To control for changes in agricultural practices introduced by the Green Revolution, I break the data in 1965, because according to the IACDS high yielding varieties of seeds were first introduced in 1966 and slowly spread over the 1970s and 1980s. Consequently, I report the findings separately for the first year of the data (1957), average productivity for the years 1957–1965, average for 1966–1987 and 1957–1987. Colonial revenues are statistically unrelated to agricultural productivity in all the specifications, providing some quantitative evidence of their randomness. Moreover, the findings on other controls such as rainfall and historical development are as expected, with higher rainfall and development both contributing to greater productivity.

Although this exercise is reassuring, it is not a conclusive test because there may be productivity differences between the post-1957 and the colonial period. Hence, I exploit another econometric strategy to address the endogeneity problem. Since

[27] The population share supported by agriculture is the omitted category in the regressions.

[28] Since the IACDS only reports data for contemporary Indian districts, the regressions exclude districts of former British India that are a part of contemporary Pakistan and Bangladesh.

[29] This includes the major crops and the following: barley, cotton, groundnut, gram, jute, other pulses, potato, ragi, tur, rapeseed and mustard, sesame, soybean, sugar, sunflower and tobacco.

Table 2
Determinants of 1911 district board educational expenditures.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Land tax revenues per-capita	0.0289***	0.0161***	0.0267***	0.0287***	0.0214***	0.0173***	0.0178***
	[0.0046]	[0.0055]	[0.0050]	[0.0074]	[0.0071]	[0.0055]	[0.0051]
Temporary settlement			0.0169***	0.0138***	-0.0034		
			[0.0043]	[0.0042]	[0.0071]		
Proportion of non-landlord				0.0146*	0.002		
(Banerjee and Iyer 2005)				[0.0076]	[0.0087]		
Province fixed effects	No	Yes	No	No	Yes	Yes	Yes
Geographic controls	No	No	No	No	No	Yes	Yes
Social and economic controls	No	No	No	No	No	No	Yes
Constant	0.0218***	0.0519***	0.0114***	0.003	0.0455***	0.0294	0.0667*
	[0.0060]	[0.0119]	[0.0043]	[0.0069]	[0.0159]	[0.0199]	[0.0392]
Observations	197	197	197	153	153	197	197
Adjusted R-squared	0.51	0.72	0.53	0.58	0.79	0.72	0.80
F-test: land tax revenues = 0	38.97	8.66	27.92	15.14	9.13	9.82	12.33
Prob > *F*	0.0000	0.0037	0.0000	0.0002	0.003	0.0020	0.0006

Notes: Robust standard errors in parentheses. *Significant at 10%; **significant at 5%; ***significant at 1%. Geographic controls include dummies for coastal districts, black, red and alluvial soil, and normal rainfall. Social and economic controls include population share supported by industry, population share supported by commerce, population share supported by professionals, population density, number of years since first railway line, income tax revenues per-capita, share of Brahmans, Christians, Buddhists, Muslims, Tribes, low castes and a measure of caste and religious fragmentation.

land revenues were revised infrequently, temporal variation in board expenditures was due to an increase in grants from provincial governments to district boards. Board expenditures are unreported in the historical documents for many of the districts in 1921, but I can still measure the effect of changes in expenditures between 1901 and 1911, for which the data are available, on changes in cohort literacy between 1911 and 1921 controlling for decadal changes in other time varying controls. This is akin to the traditional first difference estimation and effectively controls for time invariant district characteristics. Although grants were not randomly assigned, official documents state a preference for rewarding more needy (i.e. poorer) districts and those with larger minority populations. This generates a downward bias and so the first difference estimates should perhaps be viewed as a lower bound on the causal effect of expenditures on outcomes.

4. Data

For the empirical analysis, I constructed a new historical dataset merging district-level data from the colonial censuses to information reported in the Indian district gazetteers for 1911 and 1921. The dataset covers all the major provinces of British India, namely Assam, Bengal, Bihar and Orissa, Bombay, Central Provinces, Madras, Punjab and United Provinces.[30] Since comparable data on the urban centers of Bombay, Calcutta and Madras are unavailable in the gazetteers, they are excluded from the analysis. Indian districts were large administrative units with an average population of over one million in 1911. Given their size and potential heterogeneity, district level aggregation will lead to some efficiency loss relative to an analysis at a lower level of aggregation (Brown and Guinnane, 2007).[31] Nonetheless, there does appear to be significant variation across Indian districts to identify the effects of spending on literacy.[32]

The regressions use the following measures of literacy in 1911 and 1921: total literacy rate; cohort-specific literacy rate for the population aged 10–20 and 15–20; and gender-specific literacy rate. I begin the analysis in 1911 because of data constraints. A uniform definition of literacy was adopted in the 1911 census – an individual was recorded as literate if he or she could read and write a short letter to a friend – and enumerators were required to test for literacy. Officials believed the enumeration was relatively accurate in subsequent censuses relative to the pre-1911 census when Indian provinces tested for literacy in different and often inconsistent ways.[33] Other than the short reading and writing test, no formal schooling was necessary to be recorded as literate. In principle a child could become literate at home but this was perhaps true for only a small share of the population. Average literacy was low in the colonial period and the majority of literates appear to have attended some school.

Although literacy was measured more accurately beginning in 1911, the same cannot be said for the data on age enumeration. Age enumeration was plagued with inaccuracies because individuals often did not know their age.[34] This is perhaps to

[30] District boards were not constituted in some districts of Assam, Bengal, Bihar and Orissa. Consequently, they are excluded from the analysis.

[31] In general, the aggregation should make it more difficult to find statistically significant results, and it would also lead to higher R-squared relative to an individual or village level analysis.

[32] District board education expenditures in 1911 range from Rs. 0.015 to 0.34 per capita across districts, while cohort literacy among ages 10–20 ranges from 1.9 to 23 percent in the same year.

[33] Before 1911, no specific guidelines were given to enumerators to test for literacy and this led to substantial variation across provinces in the methods adopted. Although officials point to certain problems with the post-1911 enumeration such as enumerators on occasion adopting school standards, they do indicate that – the simple criterion laid down was easily understood and sensibly interpreted|| (Census of India 1921, Volume I – Report, Chapter VIII).

[34] Although ignorance was the most common reason for inaccuracy, official discussions also mention parents being superstitious about revealing the true age of an infant, young married couples overstating their age and older men (widowers and bachelors) understating their age. See Census of India 1931, Volume I, Part I – Report, Chapter IV.

Table 3
Relationship between agricultural productivity and colonial land revenues.

	Ag. productivity for major crops				Ag. productivity for all crops			
	1957	1957–1965 Avg.	1966–1987 Avg.	1957–1987 Avg.	1957	1957–1965 Avg.	1966–1987 Avg.	1957–1987 Avg.
(In 1911)								
Land tax revenues per-capita	0.0339	0.0177	0.0282	0.0249	0.0081	0.0171	0.0246	0.0223
	[0.0228]	[0.0154]	[0.0278]	[0.0235]	[0.0204]	[0.0158]	[0.0285]	[0.0240]
Population density	0.2272**	0.2242***	0.4116**	0.3563***	0.3490***	0.2340***	0.4250**	0.3654***
	[0.0979]	[0.0721]	[0.1634]	[0.1355]	[0.1101]	[0.0779]	[0.1649]	[0.1373]
Number of years since railways	0.0030**	0.0022***	0.0035**	0.0031***	0.0032**	0.0022***	0.0034**	0.0031**
	[0.0014]	[0.0007]	[0.0014]	[0.0012]	[0.0013]	[0.0008]	[0.0014]	[0.0012]
Rainfall	0.0067***	0.0050***	0.0059***	0.0056***	0.0077***	0.0051***	0.0059***	0.0056***
	[0.0019]	[0.0013]	[0.0021]	[0.0019]	[0.0020]	[0.0013]	[0.0022]	[0.0019]
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Social and economic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.1462	0.1323	0.1409	0.1359	0.111	0.1279	0.1821	0.1704
	[0.2836]	[0.1498]	[0.2536]	[0.2141]	[0.2251]	[0.1493]	[0.2633]	[0.2209]
Observations	150	150	150	150	150	150	150	150
Adjusted R-squared	0.575	0.598	0.621	0.613	0.613	0.596	0.619	0.613

Notes: robust standard errors in parentheses. *Significant at 10%; **significant at 5%; ***significant at 1%. Geographic controls include dummies for coastal districts, black, red and alluvial soil. Social and economic controls include population share supported by industry, population share supported by commerce, population share supported by professionals, income tax revenues per-capita, share of Brahmans, Christians, Buddhists, Muslims, Tribes, low castes and a measure of caste and religious fragmentation.

be expected for early 20th-century India because numeracy and literacy generally move together. The inaccuracy, however, introduces measurement error in cohort-specific literacy rates. Since uncertainty was the primary reason for incorrect responses, the measurement error is probably random (i.e. classical), and classical measurement error yields consistent estimates although the standard errors may be larger (Angrist and Krueger, 1999).

In addition to measuring literacy, I constructed measures of development such as population density, and occupational structure from the 1911 and 1921 censuses. Given the accuracy concerns of the smaller occupational categories, I focused on broad occupational types – the share of the population supported by agriculture, commerce, industry and professionals – to minimize measurement error. To control for differences in social structure across districts, I extracted information on important castes and religions, namely upper caste Brahmans who traditionally worked as priests and teachers, Muslims, Christians, Buddhists, Animistic tribes and lower castes. To identify lower castes, I used the caste lists reported in the province-specific social precedence tables of the 1901 census to calculate the number of lower castes in 1911 and 1921.[35] The castes enumerated as lower castes are the same as the castes enumerated as Scheduled Castes in post-independence India. For Punjab I used the Scheduled Caste lists from 1950 to indentify lower castes since the colonial censuses did not provide this information.[36]

Using the information on caste and religion, I also constructed a measure of caste and religious fragmentation (CRFI), similar to the ethnic-fragmentation index used in the literature. CRFI is a Herfindahl-based index equal to $1 - \sum s_i^2$, where s_i is the population share of each caste or religious group in 1911 or 1921, respectively. Similar to Banerjee and Somanathan (2007), I restricted the data to Hindu castes larger than 1 percent of the province population along with Muslims, Christians, Buddhists, Sikhs, Jains, Animistic tribes and Others that include castes that did not constitute 1 percent of the province population and small religious groups such as the Jews and Parsies.

Given the importance of geography to the computation of land revenues, I used the IACDS to construct dummy variables for black, red and alluvial soil.[37] Moreover, I extracted district-level data on normal rainfall from the 1911 census and created a dummy for coastal districts based on a visual inspection of historical maps from this period. Since proximity to railways may have influenced revenue calculations, I constructed a variable capturing the number of years between 1911 and the first year a railway line was constructed through the district.[38]

[35] The 1901 census reports social precedence tables on caste rankings. They were constructed in consultation with local people knowledgeable about the caste hierarchy. For many reasons, subsequent censuses discontinued the practice because various middle ranked castes believed their rank was higher than their reported rank and petitioned the Government to change their ranking in later censuses (Dirks, 2001). Cohn (1990), Dirks (2001), Srinivas (1996) and many other scholars have critiqued the colonial caste census, British interpretations of caste and the subsequent impact of the census on the Hindu caste system. The caste and religious data are self-reported measures, which could be mis-measured if individuals in non-upper castes changed their name to be counted as upper castes. This type of error will attenuate the estimates on the caste variables to zero.

[36] I checked the lower castes in Punjab against caste occupation data and they were associated with menial occupations common to lower castes in other provinces.

[37] Data on soil are for the post-independence period.

[38] Dave Donaldson provided the data on the district level entry of Indian railways.

Fig. 1. Map of British India and Princely States.

To obtain information on education expenditures and revenues, I referred to the Indian district gazetteers. Data on board expenditures were consistently available for 1901 and 1911, although many districts stopped reporting them after 1916.[39] For districts in the United Provinces, the gazetteers do not to report the exact land revenues collected, but instead give the revenue demand or what districts were expected to collect in specific years between 1905 and 1911. The reported year also varies by district. Since the revenue demand was fixed for 20–30 years, the inconsistency in the reporting year does not create any significant problems. There could in principle be differences between revenue demand and revenue collected, but the analysis includes geographic and economic controls that should address problems arising from such reporting differences.

In the absence of information on median income, I use per-capita income tax revenues as a proxy for income. The taxes were levied on a small share of the population such as government employees, and members of the formal non-agricultural sector with high incomes.[40] Hence, they should be interpreted as representative of the higher tail of the income distribution. Income taxes are available for the 1911 cross-section but are unavailable for Assam, Bengal, Bihar and Orissa in 1921 Fig. 1. The 1921 literacy regressions use the 1911 per-capita figures for these districts.[41]

Table 4 presents summary statistics of the variables for 1911 and 1921. Literacy was fairly stagnant between 1911 and 1921, increasing from just over 5 to 6 percent. Male literacy was higher than female, averaging 10 percent across Indian districts as compared to only 1 percent for females. Dropout, wastage and relapses into illiteracy were frequently blamed for the lack of progress in the early 20th century. Many of the occupational and development controls such as population density and commercial population also exhibit no remarkable change between 1911 and 1921. Urbanization rates increased slowly from 9.6 to 10.7 percent over the decade. In terms of social structure, upper caste Brahmans represented 5 percent of the population, lower castes comprised almost 16 percent, and fragmentation was high with a 75 percent probability that two random individuals drawn from a district would belong to different castes or religions.

39 For Madras districts, data on expenditures are reported for 1902–1903 and 1912–1913.

40 See Kumar (1982) for details of income taxes. Income tax rates were very low in the colonial period.

41 The analysis is robust if we use the 1911 per-capita income taxes for all districts in the 1921 cross-section.

Table 4
Summary statistics.

Variable	1911		1921	
	Mean	Std. dev	Mean	Std. dev
Census variables				
Overall literacy rate	5.1%	2.8%	6.0%	3.2%
Overall male literacy rate	9.3%	4.8%	10.4%	5.2%
Overall female literacy rate	0.8%	1.2%	1.2%	1.5%
Literacy rate aged 10–20	6.3%	3.6%	7.4%	4.0%
Male literacy rate aged 10–20	10.5%	5.7%	11.9%	6.0%
Female literacy rate aged 10–20	1.4%	1.9%	2.2%	2.5%
Literacy rate aged 15–20	7.4%	3.7%	9.0%	4.4%
Male literacy rate aged 15–20	12.8%	6.2%	15.0%	7.0%
English literacy	0.5%	0.5%	0.7%	0.6%
English literacy aged 10–20	0.7%	0.7%	1.0%	0.9%
Fraction urban	9.6%	8.4%	10.7%	9.8%
Population density	376	250	380	258
Fraction commerce	7.0%	3.2%	6.7%	3.1%
Fraction industry	12.3%	6.2%	11.6%	6.1%
Fraction professionals	1.6%	0.8%	1.6%	0.9%
Fraction brahman	5.0%	4.4%	5.1%	4.3%
Fraction Muslim	23.9%	26.3%	24.1%	26.2%
Fraction Christian	0.9%	2.0%	1.1%	2.2%
Fraction Tribes	3.1%	8.2%	2.8%	7.5%
Fraction low castes	16.0%	8.1%	14.8%	8.2%
Caste and religious fragmentation index	0.75	0.19	0.74	0.19
District gazetteer variables (in rupees)				
Income tax revenues per-capita	0.06	0.10	0.20	0.65
1911 District board educ. exp per-capita	0.06	0.04	.	
1911 Land tax revenues per-capita	1.46	1.03	.	
No. of observations	197	197	197	197

Sources: Government of India (1913, 1923) and Government of India (1979), Imperial District Gazetteer Series.
The dataset includes districts in the British Indian provinces of Assam, Bengal, Bombay, Central Provinces and Berar, Madras, Punjab and United Provinces with functioning district boards. Local boards were not constituted in a few districts of Bengal, Bihar and Orissa, and the hill districts of Assam. See text for more details.
Income tax revenues are missing for Garhwal district in United Provinces in 1921.

5. Results

In this section, I first present the OLS estimates, followed by the IV estimates, then the robustness checks and finally the quasi-first difference results.

5.1. Ordinary least squares

Table 5 presents OLS estimates of district board expenditures on overall literacy and cohort-specific literacy in the population aged 10–20 and 15–20. All the regressions include province fixed effects and controls for soil, rainfall, coastal districts, income, social structure, population density and railway development. Panel A focuses on the 1911 cross-section and reports the effects of 1901 expenditures on 1911 outcomes, while panel B focuses on the 1921 cross-section and uses 1911 expenditures. The controls in each cross-section are for the same year as the outcomes.

Although grants and surcharges on land revenues accounted for most of the district board income, the boards also received a small amount of public school fees as revenues. Districts with higher private returns to literacy may have also lobbied their board members to increase spending on education. A failure to control for differences in the returns to literacy can thus generate an upward bias on the OLS coefficient because districts with higher private demand will have more children attending public schools and paying fees, and higher literacy. To mitigate such concerns of omitted variables, the OLS regressions include an extensive set of controls.

The OLS findings for the 1911 and 1921 cross-sections are very similar (Table 5). A 10 percent increase in per-capita expenditures in 1901 and 1911 translates into an almost 4 percentage point increase in literacy for the 10–20 age group. The effects of public spending, on the order of one standard deviation of literacy rates, are thus economically and statistically significant. Many of the variables measuring returns such as the fraction of professionals are also positively related to literacy. Despite the controls, omitted variables related to public spending and unmeasured returns to schooling may nonetheless be driving some of the observed effects on spending. To evaluate this problem, the next section focuses on the instrumental variables strategy.

Table 5
OLS estimates of public expenditures on literacy rates.

	All ages			Cohort 10–20			Cohort 15–20		
	Total	Male	Female	Total	Male	Female	Total	Male	Female
Panel A: 1911 cross section									
District board educ ep	0.2356**	0.4094**	0.0311	0.3990***	0.6364***	0.0828*	0.3709**	0.5943**	0.0684
Per-capita in 1901	[0.0954]	[0.1801]	[0.0224]	[0.1453]	[0.2410]	[0.0443]	[0.1502]	[0.2564]	[0.0465]
Fraction professionals	0.9908***	1.6753***	0.2901***	1.1022***	1.6951***	0.4453***	1.2865***	2.0573***	0.5365***
	[0.2625]	[0.4617]	[0.0925]	[0.3696]	[0.5886]	[0.1637]	[0.3653]	[0.6234]	[0.1703]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	197	197	197	197	197	197	197	197	197
Adjusted R-squared	0.80	0.77	0.87	0.75	0.73	0.82	0.75	0.74	0.82
Panel B: 1921 cross section									
District board educ exp	0.2034***	0.3216***	0.0630**	0.3723***	0.5498***	0.1515***	0.3993***	0.6052***	0.1398***
Per-capita in 1911	[0.0566]	[0.1031]	[0.0280]	[0.0855]	[0.1298]	[0.0522]	[0.0976]	[0.1521]	[0.0522]
Fraction professionals	1.1692***	1.7879***	0.4848***	1.3191***	1.7954***	0.7086***	1.5775***	2.2691***	0.7320***
	[0.2905]	[0.5127]	[0.1304]	[0.3620]	[0.5915]	[0.1903]	[0.3963]	[0.6887]	[0.2044]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	196	196	196	196	196	196	196	196	196
Adjusted R-squared	0.78	0.75	0.84	0.76	0.73	0.83	0.75	0.73	0.81

Notes: Robust standard errors in parentheses. *Significant at 10%; **significant at 5%; ***significant at 1%. All specifications in panels A and B control for province fixed effects and include geographic controls: namely normal rainfall, dummies for black, red and alluvial soil; economic and development controls: namely population share supported by industry and commerce, population density, number of years since first railway line, income tax revenues per-capita; and social controls: namely share of Brahmans, Muslims, Christians, Buddhists, Tribes, low castes and a measure of caste and religious fragmentation.

Table 6
IV Estimates of public expenditures on literacy rates.

	All ages			Cohort 10–20			Cohort 15–20		
	Total	Male	Female	Total	Male	Female	Total	Male	Female
Panel A: 1911 cross section									
District board educ exp	0.086	0.157	−0.008	0.179	0.286	0.017	0.164	0.261	0.014
Per-capita in 1911	(0.085)	[0.157]	[0.026]	[0.127]	[0.209]	[0.052]	[0.126]	[0.214]	[0.056]
Fraction professionals	1.075***	1.811***	0.329***	1.198***	1.845***	0.494***	1.377***	2.207***	0.577***
	[0.248]	[0.438]	[0.088]	[0.350]	[0.559]	[0.160]	[0.347]	[0.587]	[0.170]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	197	197	197	197	197	197	197	197	197
Adjusted R-squared	0.80	0.77	0.86	0.75	0.73	0.82	0.75	0.73	0.82
Panel B: 1921 cross section									
District board educ exp	0.127*	0.207	0.024	0.256**	0.375**	0.091	0.261**	0.339	0.106
Per-capita in 1911	[0.076]	[0.134]	[0.034]	[0.103]	[0.163]	[0.063]	[0.130]	[0.218]	[0.071]
Fraction professionals	1.262***	1.926***	0.532***	1.460***	2.007***	0.782***	1.745***	2.591***	0.772***
	[0.269]	[0.474]	[0.122]	[0.342]	[0.557]	[0.178]	[0.374]	[0.655]	[0.194]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	196	196	196	196	196	196	196	196	196
Adjusted R-squared	0.78	0.75	0.84	0.76	0.72	0.82	0.75	0.72	0.81

Notes: Robust standard errors in parentheses. *Significant at 10%; **significant at 5%; ***significant at 1%. Regressions in panels A and B control for province fixed effects, and include the same set of controls as in Table 4 that are for the same year as the outcomes. District board educational expenditures per-capita in 1911 are instrumented using 1911 per-capita land revenues. Specification 7 in Table 2 is the first stage IV regression.

5.2. Instrumental variables results

Table 6 presents the IV results for the 1911 and 1921 cross-sections controlling for province fixed effects and other observable differences in geography, income, development and social structure across districts. 1911 per-capita expenditures are instrumented using 1911 per-capita land revenues. As seen in panel A, the IV estimates on 1911 per-capita spending are not statistically correlated with any measure of 1911 literacy. According to official sources, a child became proficient in reading and writing (i.e. literate) after completing anything from 2 to 4 years of primary school. Assuming a child begins primary school at age 5, in principle 1911 expenditures should be unrelated to 1911 outcomes. And the IV results appear to corroborate this.

The findings on 1921 literacy are reported in panel B: 1911 expenditures led to higher literacy in the 10–20 and 15–20 aged cohort that was in primary school 10 years earlier and, presumably, was the most affected 'treatment group' in 1911. The positive effects on total literacy rates in 1921 are thus due to the positive effects of spending on these 'treated' cohorts. In all the specifications, the IV estimates are smaller in magnitude as compared to the OLS estimates reported in Table 5, suggesting that the OLS coefficients are biased upwards. The magnitude of the IV estimates is still significant – a 10 percent increase in 1911 per-capita district board spending, which roughly translates into 44 additional primary schools using average 1911 district population or an 8 percent increase in school enrollment, leads to a 2.6 percentage point increase in 1921 cohort-specific literacy for the population aged 15–20.[42]

Higher expenditures can cause higher literacy by increasing access to schools, but they can also improve outcomes by moving existing pupils to better quality schools. In this view a 10 percent increase in 1911 per-capita spending would allow 4 percent of pupils to move from a low quality private school with an untrained teacher to a higher quality public school with a trained teacher.[43] While the OLS results found statistically significant effects of public spending on female literacy, the IV results indicate that higher spending did not lead to better outcomes for women. There was some substance in official arguments that more spending was unlikely to increase female literacy without parallel changes in social attitudes toward female education.[44]

5.3. Robustness checks

Colonial public investments in primary education led to substantial gains in literacy based on the IV estimates reported in Table 6. Although land revenues strongly influenced public spending on education, they could be systematically related to unmeasured returns to education, making them an invalid instrument. After controlling for objective differences in geography, income and development across districts, the exclusion restriction underlying the IV strategy is that variation in revenues was largely a function of historical idiosyncrasies uncorrelated with the error term. Since the strategy fails to identify the precise nature of the underlying variation in revenues, I subject the IV results to many robustness checks.

First, I use different cohort-specific literacy rates in 1921 as a control group because the selected cohorts were either too old or too young to be in primary school 10 years earlier. In principle, the IV estimates on 1911 public spending should not affect outcomes in these cohorts because they were not exposed to 1911 expenditures. The top panel of Table 7 (panel A) presents the IV results using 1921 cohort-specific literacy for the cohort under age 10 and cohort over age 20 as the dependent variable. Individuals in the 1921 cohort under age 10 were born after 1911 and so could not have attended primary school in 1911, and cohorts aged 20 and over were too old for primary school in 1911. As seen in panel A, the estimates on district board expenditures are statistically insignificant for the control groups.

The second strategy exploits the same idea and uses English literacy as another control outcome because public primary school expenditures should not increase English literacy unless the instrument is invalid. Primary schools in this period offered instruction in the vernacular only as compared to secondary schools where English medium instruction was more common.[45] If the unobservable factors affecting the returns to English literacy are similar to those for literacy in any other language, then English literacy offers another control outcome. The IV estimates on board expenditures should be unrelated to English literacy unless per-capita land revenues are correlated with unobservable factors that generally increase the demand for education. The first three specifications in panel B of Table 7 report the findings on total English language literacy, and disaggregated for the population aged 10–20 and 15–20. The IV estimates on spending are not positively, but in fact negatively, correlated with English literacy. Moreover, we can infer that the demand for English and non-English literacy was affected by similar forces because the share of professionals is positively related to both overall and English literacy. While this finding is reassuring, the test is not conclusive because the unobservables driving English literacy may be different from those for overall literacy.

Specifications 4, 5 and 6 in panel B of Table 7 test for more endogeneity problems. Although public spending on education leads to better outcomes, there may be some concern about how this spending relates to expenditures on other public goods that may also positively influence literacy. Rural district boards were primarily responsible for the provision of education, medical services and local infrastructure. Districts with higher land revenues would either have more money to spend on the other local services, or they could substitute spending between the different services. If areas with high public educational expenditures also spend more money on other services due to higher land revenues, then this could generate an upward bias on the IV estimates, because higher quality medical services may have an independent effect on literacy.[46] The IV estimates, however, are robust to the inclusion of 1911 medical and infrastructure expenditures as additional controls. I report only the results on cohort-specific literacy for ages 15–20 but the findings are unchanged for other measures of literacy as well.

Specification 6 in panel B, tests for another variable that may be correlated with land revenues, expenditures and literacy, namely the land tenure system. In Bengal, Bihar and Orissa, the British made landlords responsible for revenue payments (*zamindari*), whereas in other parts of the country they dealt directly with individual cultivators (*raiyatwari*) or with village bodies (*mahalwari*). Non-landlord areas had access to more public money on account of the Temporary Settlement, but they could also have a higher demand for literacy because cultivators directly negotiated with British officers. Banerjee and Iyer

[42] Data on average cost of primary schools and enrollment rates used in the calculations are for 1911-1912 obtained from the Progress of Education, Eighth Quinquennial Review (1923).

[43] Average annual expenditures for the two school types (public board and private unaided) are for 1911-12 and were obtained from the Progress of Education, Eighth Quinquennial Review (1923).

[44] Progress of Education, Quinquennial Reviews (1886–).

[45] See House of Commons (1854) and Progress of Education in India, Quinquennial Reviews (1886–1937).

[46] In theory, we could expect districts to allocate differential spending across local services, i.e. high for education and low for medical services depending on the preferences of the district. However, in the Indian context, it appears that districts with high public spending on education spent more on all other public services on average.

Table 7
Robustness checks.

	Cohort under 10			Cohort over 20		
	Total	Male	Female	Total	Male	Female
Panel A: 1921 Literacy rate						
District board educ exp	0.0283	0.0445	0.0107	0.1426	0.2468	0.0038
Per-capita in 1911	[0.0212]	[0.0320]	[0.0138]	[0.0972]	[0.1761]	[0.0363]
Fraction professionals	0.2240***	0.2787**	0.1689***	1.6449***	2.5738***	0.6275***
	[0.0774]	[0.1182]	[0.0470]	[0.3513]	[0.6454]	[0.1500]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: 1921 Literacy rate						
	English language literacy rate			Literacy rate		
	Total	10–20	15–20	Cohort 15–20		
District board educ. exp	−0.0655**	−0.0667*	−0.0937*	0.2419*	0.2516*	0.4977***
Per-capita in 1911	[0.0326]	[0.0379]	[0.0526]	[0.1336]	[0.1378]	[0.1067]
Fraction professionals	0.4224***	0.5485***	0.7030***	1.5561***	1.7216***	1.2710***
	[0.1367]	[0.1401]	[0.1559]	[0.4429]	[0.4004]	[0.4048]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
1911 Medical exp.				Yes		
1911 Local infrastructure exp.					Yes	
Proportion non-landlord (Banerjee and Iyer 2005)						Yes
Panel C: Δ lliteracy rate (1921–1911)						
	Total	Male	Female	15–20	Male 15–20	Female 15–20
Δ Educational exp	0.0649**	0.1159**	0.0142	0.1551**	0.2626***	0.0535
(1911–1901)	[0.0325]	[0.0475]	[0.0255]	[0.0664]	[0.0954]	[0.0567]
Δ Fraction professionals	0.1633	0.241	0.0857*	0.4235**	0.7877**	0.1033
	[0.0998]	[0.1672]	[0.0503]	[0.1918]	[0.3418]	[0.1090]
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *Significant at 10%; **significant at 5%; ***significant at 1%. All specifications include province fixed effects. Panels A and B include the same set of controls as in tables 4 and 5, and 1911 district board expenditures are instrumented using 1911 land revenues. In panel C the control variables are the same as in Tables 4 and 5 except that they are expressed as first differences between 1921 and 1911. See text for more details.

(2005) in an influential paper find significant differences in agricultural productivity and public investments between landlord and non-landlord districts in post-independence India. Given there is limited variation in land tenures within provinces, this should not raise any problems, but nonetheless specification 6 includes the Banerjee and Iyer (2005) variable for non-landlord districts as another control. The results are essentially unchanged although the magnitudes are larger.

While the robustness tests thus far have focused on the IV estimates, the Indian institutional context offers another estimation strategy. Although the historical sources inconsistently report expenditures in 1921, I extracted 1901 board educational expenditures for all the districts in the analysis. Since there is temporal variation in expenditures between 1901 and 1911 due to increases in grants from provincial governments to district boards, I exploit the change in spending to identify its effects on changes in literacy for the relevant cohorts. Panel C in Table 7 focuses on the production of literates, namely changes in the overall and cohort-specific literacy rate for the population aged 15–20 between 1921 and 1911. Specifications 1–3 report the findings on changes in overall literacy. Controlling for changes in other time varying controls, changes in expenditures between 1901 and 1911 are positively correlated with changes in literacy between 1921 and 1911.

The results on the production of literates in the cohort aged 15–20 are reported in specifications 4–6, which highlight that the findings on overall literacy are driven by increases in the cohort-specific literacy rates. If I had access to consistently enumerated literacy data for 1901 or perhaps expenditure data for 1921, I could run a first difference regression. In the absence of that data, these regressions are a second best option and in a sense are quasi-first difference estimations that are essentially controlling for time invariant district characteristics. The main idea is that changes in spending between 1901 and 1911 should affect outcomes for the aged 15–20 cohort between 1911 and 1921 that was in primary school between 1901 and 1911 and thus was most exposed to public spending on primary education. A 10 percent increase in spending between 1901 and 1911 leads to a 1.5 percentage point increase in cohort-specific literacy for the population aged 15–20. The magnitude of the effect is smaller than the IV estimates, but we may want to view the first difference estimates as a lower bound on the effect of public spending. Although public grants were often disbursed on a per-capita basis to districts within provinces, there was some discussion of giving more money to poorer or more backward districts that would generate a downward bias on the quasi-first difference estimates.

The findings from all the three approaches, OLS, IV and the quasi-first difference, broadly confirm the positive benefits of public spending on literacy. Given the biases associated with each estimate, we can interpret the OLS estimate as an upper bound and the quasi-first difference as a lower bound on the 'true effect'. After addressing potential endogeneity concerns, the IV and first difference estimates also suggest that public spending led to economically significant gains for male literacy without an appreciable impact on female literacy.

6. Discussion

Now that the empirical analysis has established that higher public spending in British India would have translated into better outcomes, at least for men, the next question is how we should interpret the magnitudes. Relative to the contemporary US literature on school finance reforms and test scores, where higher expenditures have either insignificant or relatively small positive effects on test scores, the estimates on colonial public expenditures may seem large.[47] But the key difference is that our outcome of interest is basic literacy that is probably more responsive to higher public spending as opposed to math test scores in the United States.

An alternate way to interpret the magnitude of the results is to use the estimates on public spending to arrive at a rough calculation of the costs associated with increasing mass literacy in British India. Even though higher public expenditures on primary education increased literacy, the colonial government may have been reluctant to increase aggregate public spending, perhaps because it was not cost-effective. A 10 percent increase in 1911 per-capita district board expenditures translates into an increase of Rs. 7107 using average 1911 district population size from the data. A 10 percent increase in expenditures leads to a 2.6 percentage point increase in 1921 literacy among the population aged 15–20, and is equivalent to making 2399 additional people, or rather males, literate. This back of the envelope calculation suggests that it would have cost the colonial government roughly Rs. 3 to make an additional person literate.[48]

Unfortunately, it is difficult to assess returns to basic literacy in the colonial period because the historical sources do not report information on wages disaggregated by education or literacy. One way to assess the cost of Rs. 3 is to consider whether a reallocation of the entire education budget toward primary education would have generated substantial increases in literacy. During the colonial period more than 50 percent of total expenditures were directed to secondary schools and colleges, and a redirection of these expenditures to primary education would have increased overall literacy by 2.4 percentage points in 1921 from, say, 7.5 to 9.9 percent.[49] Although this reflects an increase, it also emphasizes that a simple reallocation of funds within the education budget was not the way to achieve mass literacy. The colonial government would have had to substantially increase public spending to achieve significant gains.

7. Conclusion

In this paper, I explore the effects of colonial public investments on literacy across British Indian districts in the early 20th century. Using a new dataset on expenditures and outcomes, I find large and positive effects of spending on cohort-specific literacy rates using an instrumental variables strategy. A 10 percent increase in spending, or 44 additional primary schools, led to a 2.6 percentage point increase in literacy rates for the population aged 15–20. The findings are robust to a variety of falsification tests and an alternate estimation strategy akin to a first difference estimation that exploits variation in spending within districts.

The findings have three important implications for the development of education in British India. First, more public money would have led to better outcomes for male literacy. Second, the colonial educational budget in the early 20th century was insufficient to tackle the problem of mass illiteracy even if the entire budget was targeted at primary education. To increase literacy the British Government would have had to either raise taxes or reallocate revenues from other sources toward education spending. Third, higher public spending would not have had any appreciable effect on female literacy that was more responsive to demand side conditions. Social barriers against the education of women were perhaps so strong in British India that building schools was not the solution to the problem of female illiteracy.

The results also have implications beyond Indian economic history. A cursory examination of regional literacy patterns in contemporary India suggests that they are very similar to those observed in the colonial period. Cochin, which corresponds to the contemporary state of Kerala, enjoyed relatively high levels of literacy as early as 1911 (15 percent as compared to the national average of 5.6 percent) and Kerala is close to achieving universal literacy today. Western and southern India generally had above average literacy historically as compared to northern and eastern India, and this phenomenon has continued into the post-independence period. Indian districts appear to have inherited a certain educational endowment at Independence and differences in colonial investments were partially responsible for the differences in endowments. Understanding the historical circumstances and endowments is thus a necessary pre-condition for current development policies designed to remedy the poor educational performance in places such as Bihar and Orissa that unfortunately have strong and persistent historical roots.

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[47] Hanushek's (1986, 1996) evaluation of the US education literature suggests that there is a wide range of estimates but in general even at the high end of positive effects they are of small magnitude.

[48] Rs. 3 corresponds to just over 3 years of fees at an average primary school in 1911-1912.

[49] Data on 1922 expenditures on secondary schools are from the Progress of Education in India, Eighth Quinquennial Review (1917-1922).

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